AI Makes You Smarter, But None The Wiser: The Disconnect Between Performance and Metacognition

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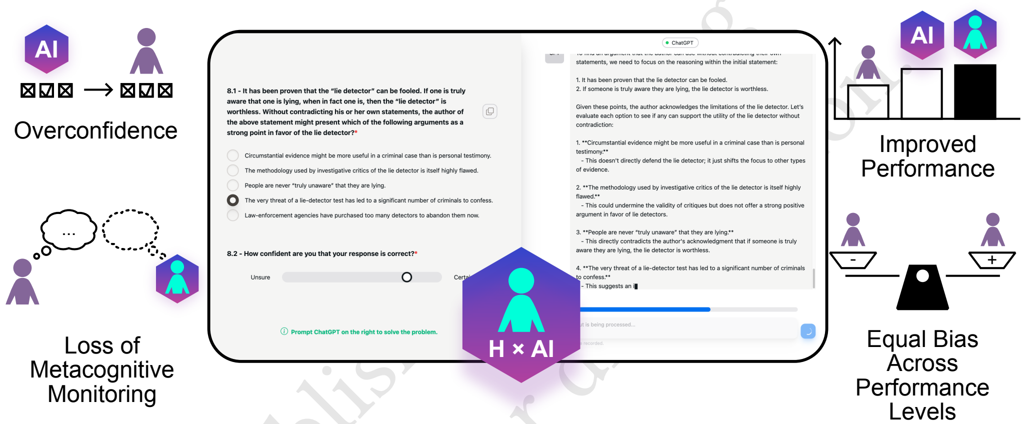


Fig. 1. Schematic visualization of our main findings. Participants using AI (ChatGPT-4o) were overconfident and overestimated their performance, although it improved compared to no AI use. The Dunning-Kruger effect was reduced with AI use.

Optimizing human-AI interaction requires users to reflect on their own performance critically. Our study examines whether people using AI to complete tasks can accurately monitor how well they perform. Participants (N = 246) used AI to solve 20 logical problems from the Law School Admission Test. While their task performance improved by three points compared to a norm population, participants overestimated their performance by four points. Interestingly, higher AI literacy was linked to less accurate self-assessment. Participants with more technical knowledge of AI were more confident but less precise in judging their own performance. Using a computational model, we explored individual differences in metacognitive accuracy and found that the Dunning-Kruger effect, usually observed in this task, ceased to exist with AI use. We discuss how AI levels our cognitive and metacognitive performance and consider the consequences of performance overestimation for designing interactive AI systems that enhance cognition.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**.

Additional Key Words and Phrases: human computer interaction

1 Introduction

Humans have always used technologies to extend their abilities [3, 18, 47]. Recent advances have aimed to improve human performance and productivity in a range of contexts [40, 64, 85, 91]. While there is evidence for an improvement in human performance with AI [5, 75, 91], these performance gains are limited by effective Human-AI Interaction (HAI). Fundamental biases, such as overtrust and overreliance, impair performance [41] up to the point that the interaction decreases overall performance as compared to having no AI at all [6]. This raises an important question of whether users can and do monitor their AI-augmented performance.

Human metacognition research investigates our ability to monitor, evaluate, and regulate our own cognitive processes [30] and, therefore, has been proposed to be essential in interactive generative AI systems [76]. Psychological research on metacognition has shown that people typically estimate themselves to be better than average [10], also called the "better-than-average effect" (see [89]).

In the context of AI, people believe AI improves performance [49], that AI predictions outperform professionals [72] and hope AI will improve their lives [15]. Research on HAI shows evidence of deficiencies in metacognitive monitoring: users are largely unaware of their performance and their performance improvement with AI. When using AI systems, users tend to overestimate its benefits, even when using a sham AI system [49, 51, 82]. However, accurate metacognitive monitoring is crucial for optimizing HAI. Inaccurate evaluation of human-AI composite performance [28] can lead to an overreliance on the system, resulting in suboptimal outcomes. From a rational perspective [61], optimal interaction with AI requires that users possess a clear understanding of their performance to adjust their behaviour.

Similarly, metacognitive judgments exhibit considerable individual variability [2, 78], which can relate to cognitive performance [78]. The Dunning-Kruger Effect (DKE) describes a cognitive bias where individuals with lower ability overestimate their competence while those with higher ability underestimate it [52]. For HAI, a DKE would suggest that low performers may not optimize their interaction with AI due to poor metacognitive monitoring. However, one could argue that if AI interaction improves overall cognitive performance by augmenting intellect [28], then metacognitive bias and its link to cognitive performance may disappear. To disambiguate these opposing arguments, we must empirically explore metacognitive monitoring, including metacognitive bias, individual metacognitive accuracy, metacognitive sensitivity, and its relation to performance (DKE) in HAI. We thus study whether AI impacts self-assessments of performance (**RQ1**), if it affects the ability to distinguish between correct and incorrect judgments (**RQ2**), and if it amplifies or reduces self-assessment bias between low- and high-performing individuals (**RQ3**).

We investigate metacognitive monitoring and the opposing perspectives on the DKE by measuring how AI affects users' metacognitive judgments. We designed an experiment where participants used AI to complete logical reasoning tasks from the Law School Admission Test (LSAT). This setting is analogous to those utilized in prior research on metacognitive abilities and the DKE [42]. By analyzing participants' self-assessments before and after AI interaction, we can directly measure the AI's influence on participants' metacognition and study its relation to task performance (DKE).

We found that while AI-assistance improves task performance in the LSAT, it also leads to overestimation of users' performance (low metacognitive accuracy). Yet, we show using a computational model that the DKE is not only smaller but disappears entirely in our sample while being present in a comparable large-scale sample without AI use [42]. Technological knowledge and critical appraisal of AI, as measured by the Scale for the assessment of non-experts' AI literacy (SNAIL) [53], increased confidence but decreased the accuracy of self-assessment.

To summarize, although AI has the potential to improve performance and level individual biases in metacognition, it carries the risk of inflated performance assessments. Our paper extends our understanding of metacognitive monitoring in HAI by investigating the interplay between metacognition, cognitive performance and AI literacy. While AI had great benefits for performance and levelled metacognitive deficits, it also brought about the bane of overestimating one's performance. We discuss how to navigate this trade-off and how to improve metacognitive accuracy to empower users to optimize their interactions with AI. Our contributions and research results are:

- Experimentally investigating the impact of AI use on metacognitive monitoring.
- Revealing that while AI can improve task performance, it leads to overestimation of performance.
- Demonstrating that the DKE is reduced when participants use AI, suggesting that AI can level cognitive and metacognitive deficits.
- Highlighting a paradox where higher AI literacy relates to less accurate self-assessment, with participants more confident yet less precise in their performance evaluations.
- Offering design recommendations for interactive AI systems to enhance metacognitive monitoring, empowering users to critically reflect on their performance.

2 Related Work

In the following section, we focus on the role of metacognitive processes in human cognition and human-AI interaction. We introduce central concepts of human metacognition and examine how self-monitoring and AI assistance can impact cognitive performance and user decision-making.

2.1 Human Metacognition

Metacognitive processes, such as monitoring or evaluating one's cognitive operations, are key for problem-solving, learning, and behavior optimization [30]. Two main constructs in research on metacognitive judgments are metacognitive accuracy and sensitivity. Metacognitive accuracy refers to how well self-evaluations align with actual performance, enabling better decision-making when limitations are recognized [20, 30]. Metacognitive sensitivity reflects the ability to distinguish between correct and incorrect judgments, often measured by confidence ratings post-decision [30]. Metacognitive accuracy is shaped by metacognitive bias (consistent over/underestimation of abilities) and noise (random fluctuations in self-assessment) [29]. Bias skews evaluations predictably, while noise introduces inconsistency, reducing sensitivity [30]. The DKE appears in the connection between metacognitive accuracy and skill. It suggests that less-skilled individuals overestimate their performance, while highly competent individuals underestimate theirs [52]. Debates exist on whether the DKE is a statistical artifact or if it reflects true population trends [34, 36]. Jansen et al. [42], as well as others (e.g., Burson et al. [13]) addressed these issues, replicating Kruger and Dunning [52] with over 3000 participants, confirming the existence of the DKE in verbal and logical reasoning tests. Low performers struggled to estimate their accuracy, while high performers slightly underestimated their performance. **Therefore, metacognitive abilities are vital for optimizing cognitive processes and relate to individual performance differences in skill, comprising the DKE.**

2.2 Metacognition in Human-AI Interaction

Augmenting human intellect has been central to HCI, as highlighted by Engelbart and others [28, 70] and metacognition has been studied, albeit seldomly quantitatively, at CHI [16, 21, 31, 37, 55, 56, 66, 71, 76, 77, 90]. Recent AI advancements have extended this vision, enabling the processing of multi-modal data—sound, visuals, and text—to solve complex problems across fields like medical treatment [59], drug discovery [58], and climate change [43], as well as in personal contexts [24]. Studies have measured AI's benefits to human cognitive performance, such as Noy and Zhang [60] finding a 40% increase in students' essay-writing speed with AI, and Bastani et al. [6] reporting a 48% improvement in math performance with ChatGPT. However, students using ChatGPT still performed 17% worse in exams compared to a control group [6], indicating that AI can sometimes hinder skill development [6, 50, 67]. These issues likely stem from suboptimal interfaces that fail to support metacognition [49, 51, 76]. Research shows that users often overestimate their AI-assisted performance and struggle to monitor or plan interactions effectively [9, 49, 51, 82]. For instance, Zamfirescu-Pereira et al. [88] found that users have difficulty crafting effective prompts, while Dang et al. [22] noted challenges in switching between tasks and writing prompts. Furthermore, explanations from AI systems are often uninformative, ignored, or lead to cognitive biases themselves [8, 27, 81, 86]. AI literacy, which involves understanding AI concepts and evaluating outputs critically, is also essential for effective interaction [53]. However, its influence on metacognitive judgements in AI-assisted decision-making and interaction optimization is unclear. Therefore, a key limitation in human-AI interaction is primarily metacognitive, involving difficulties in planning, monitoring, and evaluating interactions. **Theoretically, AI imposes new metacognitive demands on users [76], requiring enhanced monitoring and control. However, empirical research on metacognitive monitoring is scarce.**

3 Method

In the following, we motivate and document our methodological choices when conducting the study. The research software, the analysis with all associated measures and data, can be found at [blinded for review].

3.1 Participants

To explore individual differences in cognitive and metacognitive performance, we recruited a larger sample than typical DKE studies, allowing us to detect differences in metacognitive accuracy across high and low performers [26, 35][1].

We recruited 274 English-fluent participants located in the USA through Prolific. We included an attention check, requiring participants to read a short description of the study and task. They then answered two multiple-choice questions, one about the topic (logical reasoning) and another regarding which option to choose when solving the problems (the best one). We excluded thirteen (13) participants due to failing the attention check, as well as two (2) due to erroneous responses (e.g., exceeded the number of possible correct answers in estimating performance) and thirteen (13) due to low completion times.

We further analyzed data from 246 participants (identified as female 114; identified as male 130, identified as non-binary 2; Age: M = 39.85, SD = 14.53). When asked to estimate their English fluency, 218 participants reported themselves as native English speakers, 25 as fully fluent, two as conversationally fluent and one as understanding basic English. No participants preferred not to disclose their language proficiency. 14 participants in our sample reported their highest educational degree to be a doctoral degree, 58 a higher tertiary education degree (Master's level), 95 a lower tertiary education degree (Bachelor's level), 52 an upper secondary school / high school and 27 a vocational college degree. 13 participants had taken the LSAT before; their performance was slightly lower, M = 12.38, SD = 3.22, compared to those who have not taken it, M = 13.01, SD = 2.85, thus they were not excluded from the sample. We collected informed consent from each participant before the study in accordance with the Declaration of Helsinki guidelines [4]. Each participant was compensated >anonymized< upon completion.

[1]We powered for the smallest effect of interest which is the DKE. For power analysis, we used bootstrapped samples of Jansen et al. [42] with sample sizes ranging from 80 to 400 to assess the ability to detect the Dunning-Kruger effect through t-tests across quartiles. We computed the proportion of p-values < .05 to determine the optimal sample size for sufficient statistical power (80%). With this, we found that a sample size of 250-300 participants is optimal for reliably detecting differences between the upper and lower quartiles in metacognitive accuracy.

3.2 Materials

Participants' logical reasoning ability was measured with the 20 multiple-choice logical reasoning problems used by Jansen et al. [42] to approximate the Law School Admission Test (LSAT), a widely recognized, real-world assessment used in high-stakes decision-making, such as law school admissions [74, 84]. It also serves as a benchmark in machine learning research, making it ideal for comparing AI-assisted performance [45]. Using the same items enabled us to compare our results to a representative sample of participants who had no AI-assistance in the task and replicate the results of the original study by Kruger and Dunning [52].

In addition to participants' actual logical reasoning performance with AI use, perceived performance with and without AI, as well as the AI system's performance on its own, was measured using the items presented in Table 1. Lastly, we measured participants' AI literacy using the SNAIL [53] at the end of the study. The scale features 31 items to assess participants' technical understanding, critical appraisal and practical application of AI systems. The scores can be found at the end of Table 1.

3.3 Task

Participants completed 20 LSAT items in a randomized order, with the problem displayed on the left-hand side of the screen and a ChatGPT interface on the right. For each problem, participants were asked to rate their confidence in their response ("How confident are you that your response is correct?"; from "unsure" to "certain" on a 100-step slider). To ensure AI use, participants were required to prompt ChatGPT at least once, wait for its response, answer the problem and indicate their confidence. Unlimited text chat interaction was enabled during the task.

3.4 Apparatus

We inspected the software and flow of Jansen et al. [42] and carefully replicated it, integrating a side-by-side view of ChatGPT and survey interface (Figure 2). We used ChatGPT-4o (`gpt-4o-2024-05-13`) due to its widespread use in enhancing cognitive task performance [6, 24, 25]. A custom interface (Figure 2) was built to log user interactions, enabling qualitative chat analysis. The application automatically collected survey responses and chat logs and recorded them at the end of the study. We included a button to copy each problem and its answer options to the clipboard so they could be easily pasted in the chat.

3.5 Procedure

Upon entering the study on Prolific, participants were redirected to our application. After consenting to participate, they were asked to complete the expectations questionnaire. Then, they were briefly introduced to the task and allowed to test the chat interaction.

After solving the problems in a randomized order (3.3), participants were again asked to complete the expectations questionnaire, this time in the past tense. They also responded to the SNAIL questionnaire [53] and filled in their demographic information, including age, gender, occupation, education level, English proficiency, and whether they had taken the LSAT before. The study took, on average, 42 minutes to complete.

4 Results

We analyzed the data in five steps. First, we compared our sample performance to that of Jansen et al. [42]. Second, we analyzed metacognitive accuracy – the difference between objective and estimated performance – and confidence

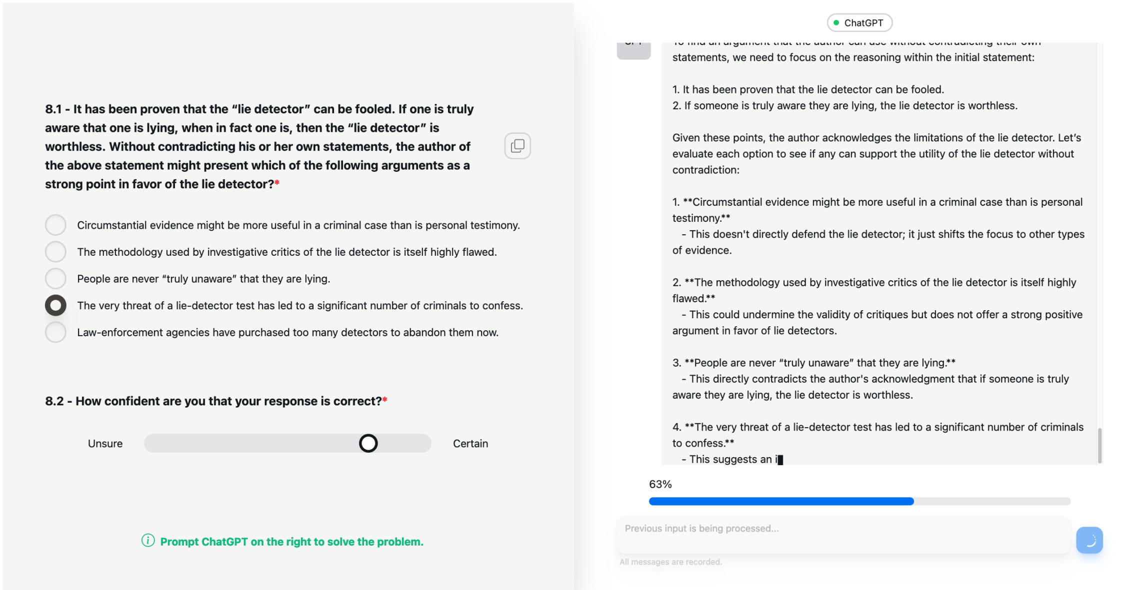


Fig. 2. Our online study application featured a horizontally split interface, with survey items and logical reasoning problems presented on the left and a ChatGPT interface on the right.

ratings to assess metacognitive sensitivity. We then correlated metacognition metrics with performance and AI literacy measures. Fourth, we used a computational model to compare our sample and Jansen et al. [42] to estimate how AI affects the DKE. Lastly, we qualitatively analyzed participant strategies [2].

4.1 Human-AI Composite Performance

To see whether there is a synergy effect of using ChatGPT in the LSAT, we compared the average ChatGPT performance (100 runs at $M = 13.65$;) to our user's average performance ($M = 12.98$, $SD = 2.88$). We find a significant difference, with participants performing slightly worse than ChatGPT alone $t(245) = -3.66, p < .001, d = -0.23$ in the task. Next, we compared our sample's performance to Jansen et al. [42] representative sample of 3543 participants, who completed the same task without any assistance ($M = 9.45$, $SD = 3.59$). We find that in our sample, participants performed significantly better with ChatGPT assistance $t(245) = 19.23, p < .001, d = 1.23$ as compared to the Jansen et al. [42] sample.

Note that the difference between our sample's and ChatGPT's performance is rather small, at less than one point. 55.28% (136 of 246) of our participants performed better than ChatGPT. However, 89.43% (220 of 246) in our sample performed better than the average score of the Jansen et al. [42] sample (see also Figure 3).

Therefore, while overall performance increased with the use of ChatGPT, the composite performance of ChatGPT and the participant overtook the performance of ChatGPT alone for slightly more than half of the participants in our sample. With our prototype's ability to enhance performance established, we can now turn our focus to investigating metacognitive abilities.

[2]Given the large samples in our study and the ceiling effects encountered Figure 5a, we do not test for the normality of residuals of our variables. Instead, we model the data using a final Bayesian computational framework, which allows for more flexible assumptions and can account for ceiling effects in our main analysis. This approach provides more robust estimates by incorporating uncertainty in a probabilistic manner, rather than relying on strict parametric assumptions.

Table 1. Descriptive Statistics for all subjective variables for the full sample and split by performance quartile (Q).

	Full sample	$Q1$	$Q2$	$Q3$	$Q4$
n	246	110	59	55	22
Performance	12.98(2.88)	10.82(3.07)	14(−)	15(−)	16(−)
Estimate	16.50(3.71)	15.34(4.31)	17.39(2.93)	17.40(3.07)	17.68(2.17)
Compared to other participants in this study, how would you rate your general logical reasoning ability when using the help of AI? (% rank)	68.08(19.3)	66.02(20.51)	70.71(18.35)	68.22(17.87)	71.0(19)
Using the AI, how many of the 20 logical reasoning problems do you think you will solve correctly?	15.96(3.63)	15.49(3.83)	16.17(3.6)	16.33(3.68)	16.82(2.13)
Without AI use, how many of the 20 logical reasoning problems do you think you would solve correctly?	11.64(4.53)	11.25(4.95)	11.36(4.19)	12.24(4.31)	12.91(3.58)
Compared to other AI systems, how would you estimate the AI system's logical reasoning ability? (% rank)	70.02(17.91)	69.04(18.04)	69.44(18.01)	70.0(18.27)	76.59(15.79)
On its own, how many of the 20 logical reasoning problems do you think the AI would solve correctly?	16.54(7.75)	15.75(4.53)	16.47(3.74)	18.31(14.42)	16.32(3.26)
Compared to other participants in this study, how well do you think you will do? (% rank)	66.95(19.84)	64.0(21.93)	69.31(17.49)	68.8(17.54)	70.77(19.3)
How difficult is solving logical reasoning problems for you?	5.38(2.01)	5.43(2.05)	5.88(2.03)	4.93(1.86)	4.91(1.93)
How difficult is solving logical reasoning problems for the average participant?	6.09(1.56)	6.14(1.62)	6.34(1.59)	5.93(1.53)	5.64(1.14)
Compared to other participants in this study, how would you rate your general logical reasoning ability when using the help of AI? (% rank)	69.83(21.86)	66.86(24.44)	72.9(19)	72.2(20.18)	70.45(18.5)
Using the AI, how many of the 20 logical reasoning problems do you think you solved correctly?	16.5(3.72)	15.35(4.31)	17.39(2.93)	17.4(3.07)	17.68(2.17)
Without AI use, how many of the 20 logical reasoning problems do you think you would have solved correctly?	11.61(4.52)	11.21(4.6)	11.78(4.47)	12.00(4.94)	12.18(3.03)
Compared to other AI systems, how would you estimate the AI system's logical reasoning ability? (% rank)	76.29(18.42)	74.04(19.51)	78.41(17.03)	77.2(19.57)	79.64(11.92)
On its own, how many of the 20 logical reasoning problems do you think the AI would have solved correctly?	17.74(8.65)	17.25(10.4)	17.68(2.84)	18.69(10.45)	18.0(2.16)
Compared to other participants in this study, how well do you think you performed? (% rank)	68.63(21.39)	65.11(23.81)	71.14(19.1)	72.13(19.21)	70.82(17.9)
How difficult was solving these logical reasoning problems for you?	5.67(2.33)	5.85(2.35)	5.64(2.24)	5.47(2.39)	5.32(2.36)
How difficult was solving these logical reasoning problems for the average participant?	6.11(2.09)	6.3(2.14)	6.07(1.95)	5.82(2.15)	6.0(2.09)
SNAIL: Technical Understanding	3.83(1.6)	3.75(1.57)	3.99(1.62)	3.77(1.66)	3.94(1.55)
SNAIL: Critical Appraisal	5.03(1.28)	5.02(1.29)	5.05(1.25)	5.01(1.33)	5.05(1.29)
SNAIL: Practical Application	5.02(1.31)	5.0(1.39)	4.99(1.27)	5.02(1.32)	5.17(1.07)

Note: M (SD) and the sample size (n). Scale for the assessment of non-experts' AI literacy (SNAIL). According to task context, rank % instructions are scaled as follows: marking 90% means you perform better than 90% of participants, marking 10% means you perform better than only 10% of participants, and marking 50% means you will perform better than half of the participants.

4.2 Metacognitive accuracy and sensitivity

Participants were inaccurate in assessing their performance after task completion. On average, they estimated solving about 17 out of 20 items (M = 16.50, SD = 3.72). This overestimation of about 4 points could be distinguished from 0, $t(256) = 14.98, p < .001, d = 0.93$. Mean confidence (rated on a scale from 0-100) for correct answers was 82.49 (14.24) and for incorrect answers 77.00 (16.52), t(244) = 8.21, p < .001, d = 0.52. To evaluate sensitivity, we conducted a Receiver

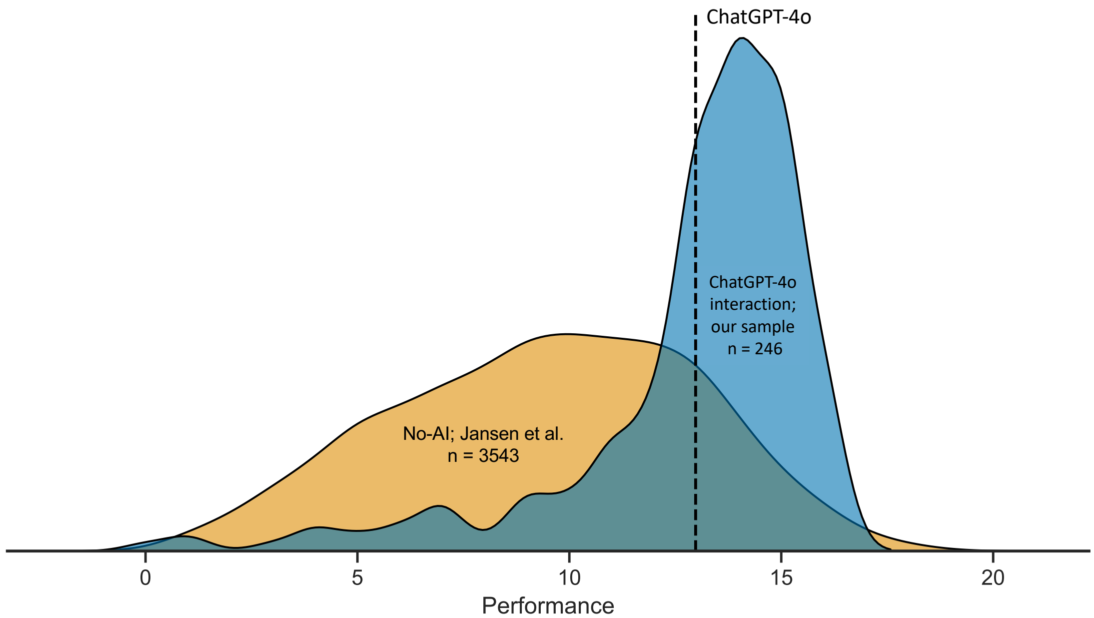


Fig. 3. Comparison of performance scores between participants interacting with ChatGPT and a dataset without AI-assistance [42]. The blue density curve represents the distribution of performance scores in a sample of 246 participants using ChatGPT, showing a peak in performance at around 14 points. The yellow density curve corresponds to a larger sample (n = 3543) from the Jansen et al. [42] dataset, with lower overall performance scores. The vertical dashed line indicates the mean performance score in the ChatGPT simulation.

Operating Characteristic (ROC) analysis. ROC analysis is a technique used to assess the performance of a judgment by plotting the true positive rate against the false positive rate across different thresholds. Here, we applied ROC analysis to understand how well participants' confidence scores predicted whether their responses were correct (see Figure 4A). The effectiveness of the confidence scores for each participant in predicting correctness can be summarized as the Area Under the Curve (AUC). A higher AUC value indicates that the confidence scores are more reliable in reflecting the participants' correctness. While most participants' (210 out of 246; 85.37%) metacognitive AUC values are above .50 (random guessing), some participants' confidence ratings were not able to distinguish between correct or incorrect trials (for the distribution of AUC values, refer to Figure 4B).

4.3 Correlation of metacognitive ability, performance, and AI literacy (SNAIL)

A number of significant relationships were found when correlating several metacognitive indices with LSAT performance and AI literacy. We observed a strong positive relation between performance and participants' average confidence estimates. Participants who perform well are also more confident on average. However, those who are, on average, more confident also overestimate their performance due to increased metacognitive bias. This is probably due to the relationship between SNAIL factors and performance estimates, where participants who express more technical knowledge and more critical appraisal also estimate their performance to be relatively higher. However, those with high technical understanding are also less accurate in their metacognitive judgements. All SNAIL factors correlated

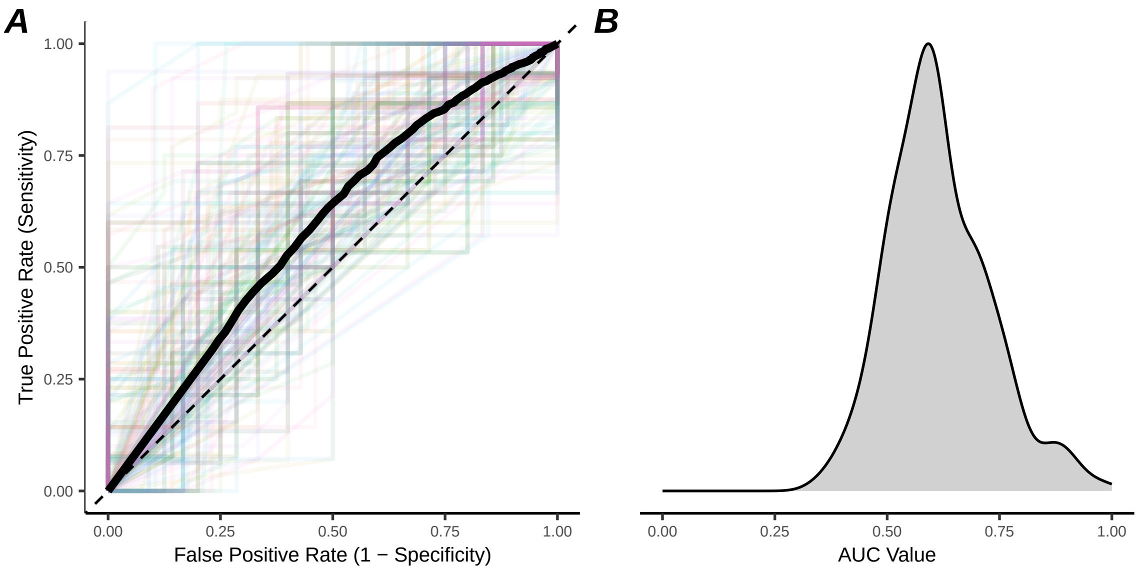


Fig. 4. (A) Receiver Operating Characteristic (ROC) curves showing the relationship between True Positive Rate (Sensitivity) and False Positive Rate (1 – Specificity) for participants (color) and pooled across participants (black). The dashed diagonal line indicates the line of no discrimination (random guessing). For each participant, we generated a ROC curve, colored lines, as well as one from pooled responses (black line), which illustrates the trade-off between the true positive rate (i.e., the proportion of correct judgments identified as correct) and the false positive rate (i.e., the proportion of incorrect judgments identified as correct) across various confidence thresholds. (B) Distribution of Area Under the Curve (AUC) values across all participants, with a peak around 0.6, suggesting variability in metacognitive sensitivity, with most participants performing above chance level (AUC = .5).

positively to average confidence. Note that these correlations are rather small and should thus be interpreted with caution. Metacognitive sensitivity (AUC and $\Delta conf$) was not related to AI literacy, performance or metacognitive accuracy.

4.4 AI use cancels the Dunning-Kruger effect

The correlation between estimated performance and actual performance is small to medium-sized (see Table 2, and for visual representation Figure 5a). While some participants were very accurate in estimating their performance, some participants were considerably off in their estimates (Figure 5a). This suggests the possibility of a DKE-like pattern, where ability in a task is related to the metacognitive ability to judge one's task performance. For the classical quantile plot, refer to Figure 5b[3].

We calculated the difference between quantiles to test whether metacognitive accuracy was worse in the low-scoring quantile than in the high-scoring quantile. Both quantile's metacognitive accuracy differed from 0, (Q1: $t(109) = -10.15, p < .001, d = -0.97$, Q4: $t(21) = -3.64, p = .002, d = -0.78$), probably due to the overall bias. However, we found that the difference for Q1 is larger than for Q4, $t(130) = 2.79, p = .006, d = 0.49$ (see also Figure 5b and Table 1). Note that this pattern of effect could be driven by metacognitive bias alone. To establish a DKE, we must first quantify the metacognitive noise in our sample. To do so, we employ a Bayesian computational model [4].

[3]Note, however, that this plot can be misleading [36]

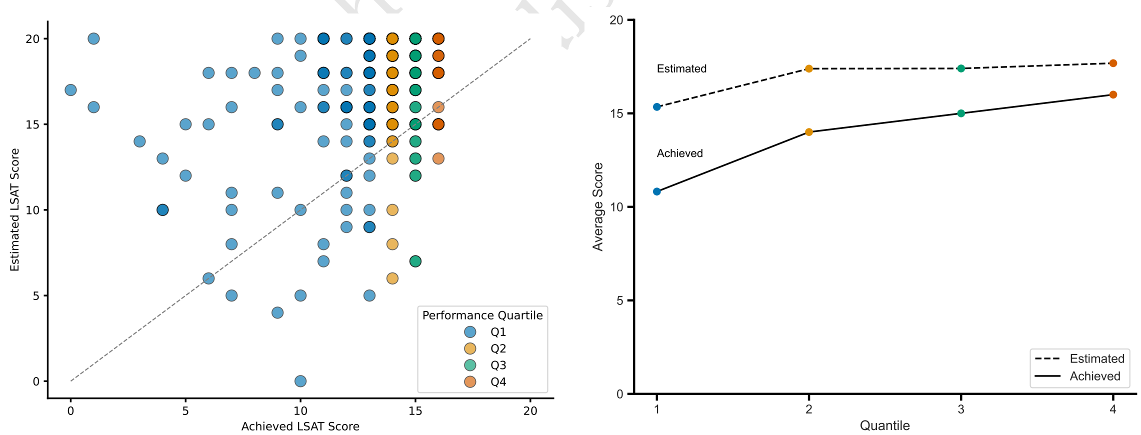
[4]For a guide on Bayesian techniques, see [12, 23, 46, 69, 80], we used the tutorial of Nathaniel Haine's as a starting point for our modelling efforts: http://haines-lab.com/post/2021-01-10-modeling-classic-effects-dunning-kruger/

Table 2. Correlation Table of Metacognitive measures and AI literacy as measured by the SNAIL

	ΔEP	Estimate	Performance	$\Delta conf$	$\mu conf$	AUC	SNAIL TU	SNAIL CA	SNAIL PA
ΔEP									
Estimate	0.72***								
Performance	-0.43***	0.32***							
$\Delta conf$	0.04	0.03	-0.01						
$\mu conf$	0.24***	0.46***	0.27***	0.10					
AUC	0.03	0.05	0.03	-0.59***	-0.08				
SNAIL TU	0.21**	0.17**	-0.06	0.12	0.13*	-0.10			
SNAIL CA	0.10	0.14*	0.06	-0.04	0.25***	0.03	0.49***		
SNAIL PA	0.05	0.10	0.06	-0.01	0.24***	0.05	0.57***	0.81***	

Note. $df = 243$, * indicates $p < .05$,** indicates $p < .01$ and *** indicates $p < .001$. ΔEP represents the difference between performance and estimated performance (metacognitive accuracy). Performance refers to the achieved task performance. $\Delta conf$ is the difference between predicted and actual confidence, while $\mu conf$ is the mean confidence (average confidence ratings). AUC refers to Area Under the Curve, with a higher AUC value indicating a more reliable confidence score in reflecting participants' correctness; SNAIL TU stands for the Technical Understanding score, SNAIL CA represents the Critical Appraisal score, and SNAIL PA is the Practical Application score.

We developed a hierarchical Bayesian model to jointly estimate participants' objective and perceived performance while accounting for latent skill, metacognitive bias, and metacognitive noise [5]. To allow for a baseline comparison of the DKE, we modelled the data jointly with that of Jansen et al. [42], whose study did not involve AI-assistance. This approach enables us to compare the DKE in our sample, where participants used AI, with the non-AI sample of Jansen et al. [42]. The model accounts for ceiling effects in performance estimates, treating scores of 20 as censored observations. Specifically, our hierarchical Bayesian model accounts for the presence of AI (k = "AI" or "no AI") in



(a) Scatter-plot of Estimated LSAT Scores as a function of Achieved LSAT Scores

(b) Classical Dunning-Kruger Quartile-plot. Comparison of Mean Estimated and Mean Achieved Scores Across Quartiles.

Fig. 5. Correlation of estimated and achieved LSAT score from different perspectives, individual Figure 5a vs. quartile-level Figure 5b.

[5]For a theoretical model, see Burson et al. [13]

estimating both the participants' achieved performance and their estimated performance. It also integrates latent skill, metacognitive bias, and group-level metacognitive noise, which scales bias and latent skill.

Let θ_i represent the relative latent skill for participant i with the prior $\theta_i \sim \mathcal{N}(0, 2)$.

The objective performance $y_{\text{obj},i}$ is modeled as:

$$y_{\text{obj},i} \sim \text{Binomial}\left(n_{\text{obj}}, \Phi_{\text{approx}}(\theta_i)\right),$$

where n_{obj} is the total number of items and $\Phi_{\text{approx}}(\cdot)$ is the approximation for the cumulative standard normal distribution.

Perceived performance $y_{\text{per},i}$ is influenced by group level bias b_k, latent skill θ_i and noise σ_k, which scales the difference of bias b_k and latent skill θ_i:

$$y_{\text{per},i} \sim \begin{cases} \text{Binomial}\left(n_{\text{per}}, \Phi_{\text{approx}}\left(\frac{\theta_i + b_k}{\sigma_k}\right)\right), & y_{\text{per},i} < n_{\text{per}}, \\ \text{Censored-Binomial}\left(n_{\text{per}}, \Phi_{\text{approx}}\left(\frac{\theta_i + b_k}{\sigma_k}\right)\right), & y_{\text{per},i} = n_{\text{per}}. \end{cases}$$

Here, n_{per} is the total number of perceived items. The priors for bias b_k and noise σ_k are the following:

$$b_k \sim \mathcal{N}(0, 2), \quad \sigma_k \sim \text{LogNormal}(0, 2).$$

Our model mitigates the issue around regression to the mean by explicitly modeling latent skill θ_i as a continuous variable with a flexible distribution. By incorporating noise σ_k that scales skill level θ_i and bias b_k, the model allows for greater variability in judgment among low-skill participants. This scaling effect, coupled with a bias term b_k and hierarchical priors, reduces the tendency for all participants to regress toward a single mean. For a DKE to exist, $b_k > 0$ and $\sigma_k > 1$ has to be satisfied. If only metacognitive bias is driving a DKE pattern, then σ_k will be centered at 1 (values of σ_k under 1 and close to zero would mean that high-performers would be less accurate in estimating their performance [6]). The bias parameter for our AI-augmented sample, b_{AI}, showed a median of 0.45 (95% HDI [0.32, 0.60]). The consistently positive bias indicates that individuals, when AI-augmented, tend to overestimate their abilities. We also find a bias b_{noAI} for the non-AI-augmented group of Jansen et al. [42] (Median = 0.23 (95% HDI [0.21, 0.25], p_b = 0.0%) although when comparing posterior samples (see Figure 6), 99% of posterior samples were larger in the AI-augmented group as compared to the non-augmented sample.

To understand how metacognitive noise affected self-assessment, we can turn to σ_k. For the non-AI-augmented group, we find a σ_{noAI} above 1, indicating noise affecting self-assessment. This group had a median of 1.78 (95% HDI [1.69, 1.88], p_b = 0.0%, see Figure 6B), indicating noise in judgment for the sample of Jansen et al. [42]. Combined with bias, this contributes to the DKE (see also Figure 6C for posterior predictions from the model). In comparison, our sample, which used AI to complete the task, was not affected by the DKE (Median = 1.01, 95% HDI [0.84, 1.19], p_b = 45.66%). Given the non-overlapping distributions (0% overlap) and the small HDI's, we can assert that with σ_{AI} being over 1, scaling of the equation of bias and skill is not present in our sample. Hence, when augmented with AI, we observe no DKE (see again Figure 6C).

[6]To fit our data into the model, we used the STAN-sampler [14]. Four Hamilton-Monte-Carlo chains were computed, each with 15,000 iterations and a 30% warm-up. Trace plots of the Markov-chain Monte-Carlo permutations were inspected for divergent transitions and autocorrelation, and we checked for local convergence. All Rubin-Gelman statistics [32] were well below 1.1, and the Effective Sampling Size was over 1000.
We then analyzed the posterior of the model. To investigate a parameter's distinguishability from zero, we utilized p_b, which resembles the classical p-value but quantifies the effect's likelihood of being zero (for b) and one (for σ) or opposite [39, 73]. Effects with $p_b \leq 2.5\%$ were deemed distinguishable. We also calculated the 95% High-Density Interval (HDI) for each model parameter; for visualization of prior and posterior, see Figure 6.

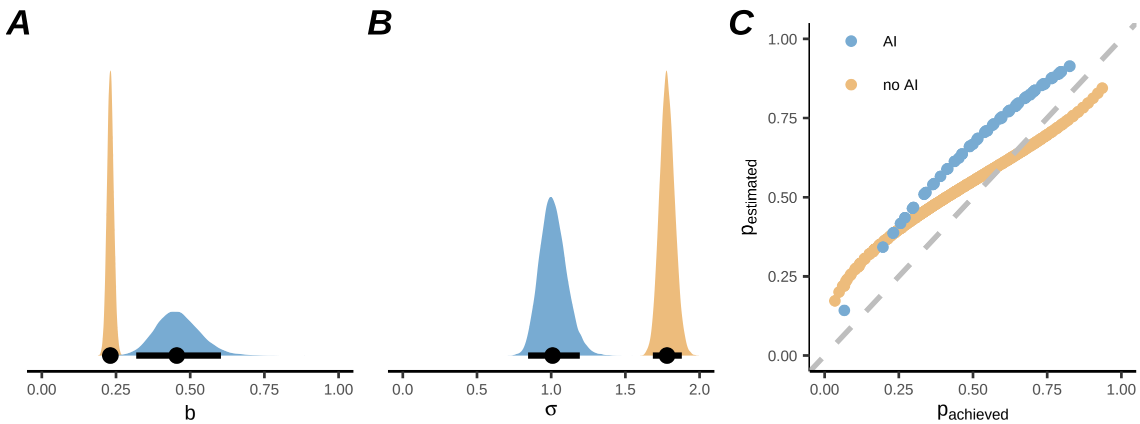


Fig. 6. Comparison posterior distributions with median and 95% HDI for the model parameters b for bias in each group for bias (Plot A) and σ (Plot B). The posterior distributions of the AI group (in blue) and no AI group (in yellow). Plot C shows the average posterior predicted values for percent correct achieved (x-axis) and percent correct expected (y-axis) for each group. The s-shape around ideal metacognitive accuracy (grey line) indicates a DKE with low-performers overestimating their performance more than high-performers (yellow; no AI group).

4.5 Qualitative Data

In addition to quantitative measurements, we analyzed the qualitative data collected during the study using an inductive thematic approach [19]. This included both the prompts participants entered into the AI chatbot interface and their responses to an open-ended question at the end of the questionnaire. The prompts were filtered to exclude those that were a direct copy from the task, ensuring that only meaningful interactions were kept for this analysis. The remaining prompts were then inductively analyzed to identify recurring themes (section 4.5.1). Responses to the open-ended question were analyzed to explore differences in AI perception during the interactions, where recurrent themes were found (section 4.5.2).

4.5.1 Analysis of prompts. In our study, we collected 11944 prompts from 246 participants, each answering 20 logical reasoning questions using the AI chatbot. To ensure quality of the data, we employed a filtering process to exclude prompts that appeared to be direct copies of questions and options verbatim, identifying these through cosine similarity comparisons with the original questions. Prompts with a cosine similarity level exceeding 95% were likely to be full copies of both the questions and options. These prompts were considered less informative and, therefore, were excluded. Prompts with cosine similarity levels between 60% and 95% likely included only the questions without the options, being also removed.

After filtering, the dataset was reduced from 11944 to 3074 distinct prompts, encompassing a variety of interactions with the AI such as salutations, queries about specific response options, requests for further clarifications, assessment of AI's capabilities, inquiries regarding the model's confidence in its responses, and other task-related instructions and inquiries.

4.5.2 Analysis of open-ended questions. A qualitative analysis of the open-ended question "Please describe how you used the AI Chatbot" revealed diverse types of perceptions and interactions with AI among participants, highlighting varying degrees of user reliance, collaboration, and trust.

Table 3. Participant Approaches to AI Use

Category	**Description**	**Actual (*M* ± *SD*)**	**Perceived (*M* ± *SD*)**
High level of trust	Participants relied heavily on AI ("blindly-trusted"), copying and pasting questions without critically assessing AI's outputs or further inquiry. 58.94% (145 out of 246)	13.014 ± 3.062	16.844 ± 3.679
Collaborative Partner	Participants perceived AI as a collaborative partner rather than a mere tool, engaging in joint problem-solving and using inclusive language ("we did this") when describing their interactions. 12.60% (31 out of 246)	13.065 ± 3.176	18.161 ± 1.881
Complementary Tool	Participants used AI strictly as a complementary tool for verification of their independently formulated answers, maintaining control of the process. 21.54% (53 out of 246)	10.5 ± 3.317	16.000 ± 6.733
Inconclusive/Did Not Use	Participants either provided inconclusive responses regarding the strategy used during the experience or did not find the AI tool useful. 6.91% (17 out of 246)	13.302 ± 2.729	16.302 ± 4.012

The majority of participants demonstrated a high level of trust in AI, often accepting its suggestions without further inquiry. This behavior raises concerns about overreliance on AI, as noted by Lu and Yin [56]. 12.60% of participants perceived AI as a collaborative partner, using inclusive language and viewing it as part of a joint effort rather than just a tool. Another 21.54% of participants viewed AI strictly as a complementary tool, using it cautiously for verification while maintaining control of the problem-solving process. Finally, 6.5% either provided inconclusive responses regarding the strategy used or did not find the AI tool useful. The data reveals diverse ways participants perceived AI, providing insights into HAI dynamics and individual variability.

5 Discussion

This study offers insights into metacognitive monitoring in HAI by examining how users with varying competence interacted with AI during logical reasoning tasks. We explored the impact of AI on metacognitive accuracy, focusing on the DKE, user confidence, and AI literacy. Our findings reveal a significant inability to assess one's performance accurately when using AI equally across our sample.

5.1 Effect of AI Literacy on Metacognition in Human-AI Interaction

While AI users in our sample outperformed those in Jansen et al. [42], they consistently overestimated their performance by about four points, aligning with previous HCI research [49, 51, 82]. The moderate correlation between estimated and actual scores (Table 2), with many participants rating themselves higher than the most skilled in the sample (Figure 5a) suggest that AI improves performance but leads to highly biased self-assessments.

This disconnect between actual and perceived performance mirrors earlier findings on overtrust and overreliance in AI systems [49, 56, 72]. Overconfidence may impair users' ability to evaluate their performance without AI, posing challenges for designing balanced human-AI interfaces. The classic DKE, where lower performers overestimate and higher performers underestimate their performance, disappeared with AI use, suggesting AI levels performance but does not correct inflated self-assessments. Metacognitive bias was doubled for the entire sample compared to Jansen et al. [42].

We have also found an unexpected link between AI literacy and metacognitive accuracy. Participants with higher AI literacy were less accurate in self-assessments, contradicting the assumption that AI literacy improves metacognitive monitoring. Familiarity with AI may enhance the better-than-average effect [10, 89], leading to overestimation of both relative and absolute performance.

Metacognitive sensitivity further explains these effects. Our ROC analysis showed that while participants were generally confident, they tended to overestimate the correctness of incorrect responses, indicating low metacognitive sensitivity. This suggests participants did not adequately monitor their performance when using AI and were largely unaware of their performance post-decision.

Qualitative data revealed varied perceptions of AI's role, from a tool to a teammate, but these differences did not significantly affect performance or metacognitive accuracy, contradicting theories that suggest interaction framing impacts outcomes [63, 82]. Regardless of user perception, the core metacognitive challenges in HAI persist.

5.2 Integration into theory

For HCI, high metacognitive bias leads users to overestimate their performance and over-rely on AI systems [57], reducing their ability to critically monitor HAI outcomes [76]. When analyzing our results through Rasmussen's SRK Model [68, 87], we find that heavy AI reliance shifted participants from reflective, knowledge-based processing to rule- or skill-based processing, reducing critical self-assessment and resulting in increased confidence but lower metacognitive accuracy. From a computational rationality perspective [61], this bias may be an adaptive response to AI presence, as participants may optimize perceived utility (e.g., efficiency) rather than monitoring HAI. This is supported by low metacognitive sensitivity and participants' reliance on copy-pasting rather than higher-level metacognitive strategies (e.g., AI as collaborator). Such aligns with Villa et al. [82], who found reduced error-processing when participants believed they were using sham AI. Therefore, our study provides further evidence of diminished metacognitive monitoring in HAI.

A lack of HAI monitoring explains several effects. First, it clarifies why AI use leads to adverse learning outcomes [1, 6]; users are overly optimistic and fail to monitor evolving joint performance. Second, while AI tools offer perceived empowerment and efficiency [49, 51, 82], the lack of reflection hinders users' ability to assess real benefits (placebo effect). Additionally, it explains persistent overreliance and overtrust [48], and why AI explanations are rarely integrated into behavior [5, 33, 86]. Though psychology and HCI offer methods to improve metacognitive judgments (refer to Table 4), they may provide only temporary solutions. Rafner et al. [67] calls for a systemic, long-term strategy, considering cognitive and metacognitive deskilling risks at individual, team, and organizational levels. This emphasizes the need for strategies that foster cognitive resilience and critical engagement with AI over short-term fixes targeting immediate metacognitive deficits.

5.3 Limitations

While our study provides valuable insights into metacognitive monitoring in HAI, several limitations may affect the generalizability and interpretation of our findings. First, the apparent absence of the DKE, "being unskilled and unaware" [26], in our study may not fully reflect the underlying cognitive dynamics. AI interaction may have made participants equally skilled yet unaware of their performance rather than eliminating the DKE. AI-enhanced performance might have decoupled metacognitive judgments from cognitive performance (e.g., high confidence, low sensitivity, and high

Table 4. Issues, Consequences, and Design Principles to Address Impaired Metacognitive Monitoring in Human-AI Interaction

Metacognitive Issue	Consequences	Design Principles
Overreliance on AI outputs	Users trust AI outputs without critical assessment, leading to reduced self-reflection and failure to notice AI errors.	• Confidence calibration to align user confidence with AI output uncertainty [57] • AI uncertainty visualization to make AI output reliability transparent [7, 65] • Explanatory AI interfaces to clarify AI decision-making processes and enable users to assess validity [44]
Loss of metacognitive monitoring	Users are unable to accurately assess their own performance or monitor task progress, especially in complex decision-making tasks.	• Post-task reflection to encourage users to evaluate their performance after interacting with AI (for a starting point, see [76]) • Cognitive forcing strategies such as prompts to promote critical thinking and reduce automatic reliance on AI outputs [11]

bias). Second, we relied on LSAT questions as our logical reasoning task. While these assess logical reasoning, they may not capture the diversity of real-world reasoning skills or domains. Furthermore, these tasks might overlap with the AI's training data, limiting generalizability. However, since ChatGPT-4o's performance was imperfect (68.25%), we believe our findings still apply to other tasks. Third, our focus on LSAT-based reasoning limits the scope of metacognitive biases across domains. Future research should use diverse tasks (e.g., writing creative texts with an LLM) to examine how AI interaction affects metacognitive monitoring and whether improvements in accuracy generalize across tasks. Lastly, we found little effect of different AI strategies on metacognitive accuracy and performance. The AI's role, whether as a tool, collaborator, or teammate, did not significantly impact participants' performance evaluation. Future research should explore how different AI roles (e.g., tool vs. collaborator) influence metacognitive accuracy and task performance.

5.4 Implications

Following Van Berkel and Hornbæk [79], we identify three types of implications of our work: theoretical, design and methodological. For HCI theory, while biases in HAI have been studied [8, 38, 49, 54], little research has analyzed the metacognitive mechanisms underlying these biases and their impact on interaction [76]. Applying metacognition theory to HAI (e.g., see Colombatto and Fleming [20]), offers a new perspective for improving metacognitive monitoring in interaction design. Our findings also suggest that AI literacy alone is insufficient for optimal metacognitive abilities in HAI, challenging current political efforts like the EU AI Act [62] that prioritize AI literacy for human-centered AI interaction. For design, we propose a new HAI interaction model that integrates metacognition and new design concepts. In the short term, designers should adjust interaction models, similar to Buçinca et al. [11], and consider broader sociotechnical risks [67]. In the long term, HCI must develop a specific HAI interaction model that enhances metacognitive functions to support knowledge-based interactions [68]. Real-time sensing of metacognitive states (e.g., user agency) and adapting interfaces to cognitive states [17, 83] could be crucial for optimizing HAI. Methodologically, our study introduces a new approach to analyzing biased decision-making in repeated AI interactions. Using computational methods, researchers can distinguish general biases from those emerging in post-decision processes (e.g., increased metacognitive noise), allowing for a more precise analysis of bias development in HAI.

6 Conclusion

Our study found that participants using an LLM to complete a logical reasoning task were unable to accurately assess their own performance, often overestimating it. With AI use, the DKE was eliminated. Based on these findings, we suggest developing new interfaces for interactive AI that are designed to enhance metacognition, allowing users to monitor their performance more accurately.

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